

Project 8

Redundancy and Feasible IK Families in 7R Robots

Format: Group of two students; simulation-based project.

Course foundations: Lectures 4 and 8; Exercises 1 and 4.

Motivation

For a redundant robot, a fixed tool pose can be realized by a continuous family of postures. A selected redundant angle provides a useful way to inspect that family, but joint limits and singularities restrict it. Understanding those restrictions is necessary before choosing a redundancy schedule along a workspace path.

Task

Overall objectives. Analyze a small family of spherical-shoulder/elbow/wrist 7R manipulators and explain how geometry changes fixed-pose IK families, their classification, and feasibility over a short path.

- **Define a bounded geometric family.** Use iiwa-like geometry as a reference and vary only the upper-arm to forearm length ratio over three instructor-supplied parameter sets with the same total arm length. The parameterized FK/IK implementation will be provided. Verify the reference against its URDF and describe the synthetic variants using their actual kinematic frames rather than unchanged meshes.
- **Inspect fixed-pose self-motion.** At two selected full poses, sample $\lambda = q_3$ and evaluate all returned IK candidates. Plot joint values versus λ , check full-pose closure, and identify intervals removed by limits or loss of regularity. Explain the locally one-dimensional nature of the regular fixed-pose solution set.
- **Classify the recovered families.** Preserve the supplied analytical branch labels, document their geometric meaning where available, and track their continuation across neighboring samples. Distinguish a label within a fixed-angle slice from a connected component of the complete self-motion set. Do not use solver array order as identity.
- **Generate comparative feasibility charts.** For one common short pose path, plot path sample horizontally and redundancy angle vertically, using a separate panel for each labelled candidate family. Distinguish absent IK, pose error, joint-limit violation, full-Jacobian rank loss, and reduced-chart failure. Compare the three geometries on identical axes.
- **Interpret feasible connections.** Extract one candidate corridor or explain the sampled obstruction for each geometry. Re-solve intermediate poses before accepting neighboring chart cells as connected. Compare feasible-angle width and minimum joint-limit/regularity margins, and explain what the chosen chart can miss.

Specifics and scope

Use three geometries, two fixed poses, and one short path, initially with about 100 pose samples and 120 redundancy angles. Refine one apparent boundary or connection. This is a geometric family study, not a full trajectory-optimization project. The provided solver must support all three parameter sets; do not apply an unmodified iiwa-specific solver to a changed geometry.

For every chart, keep pose validity, joint limits, full-Jacobian regularity, and reduced-chart regularity as separate diagnostics. The rectangular 6×7 Jacobian has no determinant. A feasible pixel is not a validated edge: re-solve intermediate prescribed poses and check joint continuity within limits. If collision checking is unavailable, state that feasibility is kinematic only; a clearance proxy is not a full collision test.

Expected outcome

Understand the URDF reference, the parameterized D–H geometry, redundant IK families, and the relation between classification and feasibility. Deliver self-motion plots and branch-labelled path-sample versus redundant-angle charts that explain how the selected geometric change affects feasible connections.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The KUKA LBR iiwa 7 URDF is `lectures_main/assets/models/iiwa7/iiwa7.urdf`; STL meshes are in that directory. The accompanying `iiwa7_free_joints.urdf` is a mathematical continuation model; apply physical limits separately. The course analytical IK model will be supplied; reuse it rather than deriving a general 7R solver. The three family parameter sets and compatible parameterized IK will be supplied separately. Use Lecture 8 and Exercise 4 for chart construction; the course reference data can check the original iiwa member only.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL's 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.